Eleni Karagianni

Creative Technologist · AI and Multi-Agent Systems

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Full UK right to work, no sponsorship required · Licensed Architect-Engineer, Technical Chamber of Greece (TEE)

PROFILE

Creative technologist with a background in architectural engineering. I build multi-agent AI systems that turn large multimodal datasets (text, images, video, 3D) into interactive tools, from the first question to a working prototype: scraping and embedding pipelines, retrieval, LLM agent design and evaluation, and MCP tool use that lets agents act in live 3D scenes. My training in architecture and urban design keeps the work tied to real users and real places. In 2026 my projects won the Bartlett's Best Computation Prize and Theory Report Prize, and I was invited to teach at the school.

EXPERIENCE

Workshop Co-Lead, The Bartlett School of Architecture, UCL — Oct 2026
London, UK

"Designing with Disagreement", B-Pro two-week intensive, co-led with Ruijia Zheng

- Designed a new two-week course from scratch: curriculum, briefs, schedule and assessment.
- Wrote all teaching materials, including Jupyter notebook templates that let 23 postgraduate students, most with no coding background, build working multi-agent systems.
- Guided teams through scraping and curating topic datasets, grounding agents in that material only, and tuning personas, settings and thresholds.
- Teams deployed opposing agents that debated across text, image and 3D through a Blender MCP connection, presented school-wide with process films.

Urban Safety Research and Advisory, LOvMe.ai — 2023–2025
London, UK

Early-stage venture building an AI companion for personal safety on everyday city journeys.

- Worked on problem definition and early research behind the "Safety Fit Score", a tool for rating the safety of routes and journeys.
- Planned and ran early user research and interviews on how students and professionals experience safety on city journeys.
- Advised on how street design, lighting, activity and sightlines shape perceived and actual safety, drawing on my undergraduate thesis. Still give product feedback and join key meetings.

Architectural Intern, Urban Soul Project — 2023
Thessaloniki, Greece · full-time, 3 months

- Produced most of the drawing sets for 5+ residential and workplace projects in parallel, including interiors for PwC's Athens campus.

Architectural Designer, Angelos Koukoulis — 2023–2024
Thessaloniki, Greece · part-time, 6 months

- Adaptive reuse of commercial buildings into housing: surveys, drawings, visualisations, coordination with engineers and clients.

EDUCATION

MArch Urban Design, The Bartlett School of Architecture, UCL — 2025–2026
London, UK

Integrated Master's in Architectural Engineering, University of Thessaly — 2020–2025
Volos, Greece · 5-year professional degree

- Final grade 9.3/10, graduated top of class
- Research thesis: “Under the Night Veil: The Night Urban Experience Through the Female Lens” (9.82/10)

Certificates: DelftX, Urban Design for the Public Good: Dutch Urbanism (TU Delft, 2024) · MITx, Becoming an Entrepreneur (MIT, 2021)

AWARDS

- Best Computation Prize, The Bartlett, UCL, 2026 (Nine Acts of Ostranenie, with Ruijia Zheng)
- Theory Report Prize, The Bartlett, UCL, 2026 (The Unseen City)
- Honourable Mention and exhibition, Playing Models 2026, SCI-Arc, Los Angeles (Nine Acts of Ostranenie)
- IKY Academic Excellence Award, Greece, 2026

SKILLS

AI and agents: multi-agent orchestration (role-based agents, debate and critique loops, audit meta-agents); LLM integration and prompting (local open models via LM Studio incl. Qwen, OpenAI GPT-4o mini, Claude API); Model Context Protocol (MCP) tool use; retrieval and grounding in curated corpora; CLIP embeddings; agent evaluation

ML and data: Python, pandas, Jupyter; scikit-learn (Isolation Forest, LOF, TF-IDF, PCA); Self-Organising Maps, UMAP; image classification; audio and spectral analysis; web scraping and metadata structuring (approx. 180,000 items); graph routing (Dijkstra)

Build: HTML, CSS, JavaScript; interactive web interfaces and 3D data visualisation; Blender Python API (bpy); TouchDesigner; Git, Vercel

3D and design: Blender, Rhino, Revit, AutoCAD, SketchUp, 3ds Max, 3D printing; Adobe Creative Cloud, DaVinci Resolve

Languages: Greek (native), English (fluent)

PROJECTS

Nine Acts of Ostranenie, multi-agent design system 2026
The Bartlett, UCL · with Ruijia Zheng · Best Computation Prize · exhibited at SCI-Arc

- Co-built a nine-agent system that turns an archive of approx. 180,000 books, photos, film stills and subtitles into spatial design proposals, argued over and modelled live in a shared 3D scene.
- Data pipeline: scraping, cleaning and CLIP embeddings, then anomaly scoring (Isolation Forest, LOF) to give each pair of "normal" and "odd" agents an opposing worldview.
- Led agent roles, retrieval rules and protocols, including three meta-agents (curator, contradiction-finder, blind-spot detector) and a human-in-the-loop curator, over approx. 100 runs on a locally hosted Qwen model.
- Found that "odd" images get machine descriptions with 65% more verbs ($p < 0.001$), which shaped how the agents speak.
- Perception-to-geometry loop via Blender MCP: agents place cameras in their own models, read the geometry and reshape it. Live archive map

Night Walk, safer-route prototype 2026
Individual · King's Cross to Camden

- Scored 8,165 street segments from OpenStreetMap and police.uk data on lighting, footfall, sightlines, help nearby and recorded incidents.
- Four walkers with different needs negotiate one route home (Dijkstra per walker, minimax-regret choice), simulated over 2,560 trips to find the streets that need fixing. Live prototype

Turin Machine, language-to-form generator 2025
The Bartlett, UCL · individual

- Two LLM agents with scripted personas walk 23 Turin locations and converse, grounded in approx. 3,200 scraped films, archive videos and building records.
- Mined the dialogue with TF-IDF, PCA and SOMs into a 42-word spatial vocabulary that drives a custom Blender script; editing one word regenerates the structure. Code on GitHub

The Unseen City, research dissertation 2026
The Bartlett, UCL · Theory Report Prize

- Examined what AI systems "see" in cities and what they discard as noise, through MIT's Streetscore and my own agents.